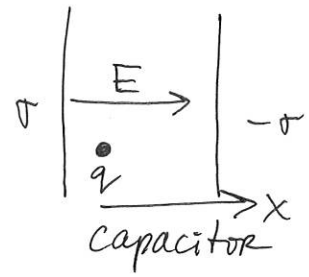
Relativistic dynamics.

Newton: $\frac{d\vec{p}}{dt} = \vec{F}$

What replaces it in relativity?

Consider stationary frame $K$




$\vec{E} = E\hat{x}$

$\frac{dp}{dt} = qE$ (in $K$) where $p = p(v)$ which should generalize $p = mv$ in Newton physics.

Next consider $K'$ moving with instantaneous velocity of the particle $V = v$, so that in $K'$ $v' = 0$.

But $dv' \neq 0$. Instead $dv' = \gamma^2 dv$ because

$$v' = \frac{v - V}{1 - \frac{vV}{c^2}}$$

$$dv' = \frac{dv}{1 - \frac{vV}{c^2}} - \frac{(v - V)}{\left(1 - \frac{vV}{c^2}\right)^2} \frac{-dv\, V}{c^2} =$$

$$= \frac{dv}{1 - \frac{vV}{c^2}} \left[ 1 + \frac{(v - V)\frac{V}{c^2}}{1 - \frac{vV}{c^2}} \right] = \frac{1 - \frac{V^2}{c^2}}{\left(1 - \frac{vV}{c^2}\right)^2} dv$$

$\Rightarrow$ for $K'$ with $V = v$ $dv' = \frac{1 - v^2/c^2}{\left(1 - \frac{v^2}{c^2}\right)^2} dv = \frac{dv}{1 - \frac{v^2}{c^2}} =$

$$= \gamma^2 dv$$

What is electric force in K?

We postulate that charge $q$ is 4-scalar, therefore invariant (this follows from experimental observations!).

$\Rightarrow q' = q$ always.

In our case $E' = E$ because $E$ is determined by charge density $\sigma$ on the plates which does not change. And element of area $dy\,dz$ does not change.

$\Rightarrow \quad E' = E.$

$\Rightarrow$ in $K'$ we have $m\dfrac{dv'}{d\tau} = qE'$

(because in $K'$ motion is not relativistic!)

Note that $d\tau$ = proper time = time in $K'$.

$$\Rightarrow \quad \frac{m\gamma^2 dv}{\frac{dt}{\gamma}} = m\gamma^3 \frac{dv}{dt} = qE \quad \text{in frame } K.$$

$\Rightarrow$ comparing $\dfrac{dp}{dt} = qE$ with $m\gamma^3 \dfrac{dv}{dt} = qE$

$$dp = m\gamma^3 dv = \frac{m\,dv}{\left(1 - v^2/c^2\right)^{3/2}}$$

$$\int_0^p dp = m\int_0^v \frac{dv}{\left(1 - v^2/c^2\right)^{3/2}} = mc\int \frac{\cos\theta\, d\theta}{\left(1 - \sin^2\theta\right)^{3/2}} =$$

$$\frac{v}{c} = \sin\theta \qquad = mc\int \frac{d\theta \cos\theta}{\cos^3\theta}$$

But $\dfrac{d\theta}{\cos^2\theta} = d(\tan\theta)$

$$\Rightarrow \quad p = mc \tan\theta = mc \frac{\sin\theta}{\cos\theta} = mc \frac{v/c}{\sqrt{1-v^2/c^2}} = \frac{mv}{\sqrt{1-v^2/c^2}}$$

Next energy: $\mathcal{E} = \frac{1}{2} m v^2$

$$d\mathcal{E} = \vec{F} \cdot d\vec{r} = \vec{F} \cdot \vec{v}\, dt = \vec{v} \cdot d\vec{p}$$

$$\Rightarrow d\mathcal{E} = \frac{mv\,dv}{\left(1 - v^2/c^2\right)^{3/2}} = \frac{1}{2} mc^2 \frac{d\left(\frac{v^2}{c^2}\right)}{\left(1 - \frac{v^2}{c^2}\right)^{3/2}} = d\left(\frac{mc^2}{\sqrt{1-v^2/c^2}}\right)$$

$$\Rightarrow \quad \mathcal{E} = \frac{mc^2}{\sqrt{1-v^2/c^2}}$$

4-vector of momentum:

$$u^\alpha = \frac{dx^\alpha}{d\tau} = \gamma(c, \vec{v}) \quad , \; u_\alpha = \gamma(c, -\vec{v})$$

Note that $u^\alpha$ is fully determined by $\vec{v}$ $\left(\gamma = \left(1 - v^2/c^2\right)^{-1/2}\right)$

4-momentum : $p^\alpha = \underset{\substack{\uparrow \\ \text{4-scalar}}}{m} u^\alpha = m\gamma(c, \vec{v})$

$$p^\alpha = \left(\frac{mc}{\sqrt{1-v^2/c^2}}, \frac{m\vec{v}}{\sqrt{1-v^2/c^2}}\right) = \left(\frac{\mathcal{E}}{c}, \frac{m\vec{v}}{\sqrt{1-v^2/c^2}}\right) = \left(\frac{\mathcal{E}}{c}, \vec{p}\right)$$

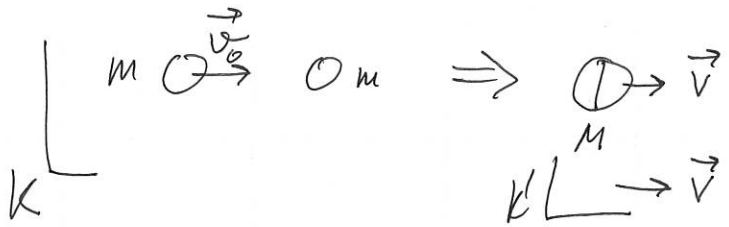
$$p_\alpha = m\gamma(c, -\vec{v}) = \left(\frac{\mathcal{E}}{c}, -\vec{p}\right)$$

Observe $p_\alpha p^\alpha = \frac{\mathcal{E}^2}{c^2} - \vec{p}^{\,2} = \frac{m^2c^2}{1-\frac{v^2}{c^2}} - \frac{m^2v^2}{1-\frac{v^2}{c^2}} = m^2c^2$

$$\Rightarrow \quad p^2 + m^2c^2 = \mathcal{E}^2/c^2 .$$

$$\begin{pmatrix} \frac{\mathcal{E}'}{c} \\ p_x' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \frac{\mathcal{E}}{c} \\ p_x \end{pmatrix} = \begin{cases} \gamma\left(\frac{\mathcal{E}}{c} - \beta p_x\right) \\ \gamma\left(p_x - \beta \frac{\mathcal{E}}{c}\right) \end{cases} \Rightarrow \begin{cases} \mathcal{E}' = \gamma(\mathcal{E} - v p_x) \\ p_x' = \gamma\left(p_x - \frac{v}{c^2}\mathcal{E}\right) \\ p_{y,z}' = p_{y,z} \end{cases}$$

The meaning of $mc^2 = \mathcal{E}_0$ = rest energy: consider inelastic collision




Let $K'$ be center-of-mass frame (CM)

$\Rightarrow$ total momentum in $K'$ $p_{tot} = 0$.

$$\Rightarrow \sum p_x' = \gamma\left(\sum p_x - v \sum \frac{\mathcal{E}}{c^2}\right) = 0$$

$$\Rightarrow \quad v_{CM} = \frac{c^2 \sum p_x}{\sum \mathcal{E}}$$

$$\vec{v}_{CM} = \frac{c^2 \sum \vec{p}}{\sum \mathcal{E}} \xrightarrow[v \ll c]{} \frac{c^2 \sum \vec{p}}{\sum mc^2} = \frac{\sum \vec{p}}{\sum m}.$$

In $K'$ combined object $M$ is at rest.

It moves with $\vec{v}_{CM}$ in $K$

$$\vec{v}_{CM} = c^2 \frac{m\gamma_0 \vec{v}_0}{mc^2\gamma_0 + mc^2} = \frac{\gamma_0 \vec{v}_0}{1+\gamma_0} = \frac{\vec{v}_0}{1+\sqrt{1-\frac{v_0^2}{c^2}}}$$

$\leftarrow$ momentum of $m$

$\uparrow$ rest energy of $m$

Conservation of momentum in K:

$$m\gamma_0 \vec{v}_0 = \frac{M\vec{V}}{\sqrt{1-V^2/c^2}} = \frac{M\vec{v}_0}{\left(1+\sqrt{1-\frac{v_0^2}{c^2}}\right)\sqrt{1-\left(\frac{v_0/c}{1+\sqrt{1-\frac{v_0^2}{c^2}}}\right)^2}} =$$

$$= \frac{M\vec{v}_0}{\sqrt{\left(1+\sqrt{1-\frac{v_0^2}{c^2}}\right)^2 - \frac{v_0^2}{c^2}}} = \frac{M\vec{v}_0}{\left[1+2\sqrt{1-\frac{v_0^2}{c^2}}+1-\frac{v_0^2}{c^2}-\frac{v_0^2}{c^2}\right]^{1/2}}$$

$$= \frac{M\vec{v}_0}{\sqrt{2\sqrt{1-\frac{v_0^2}{c^2}}+2\left(1-\frac{v_0^2}{c^2}\right)}} = \frac{M\vec{v}_0}{\sqrt{\frac{2}{\gamma_0}+\frac{2}{\gamma_0^2}}} = \frac{M\gamma_0\vec{v}_0}{\sqrt{2(1+\gamma_0)}}$$

Thus $M = m\sqrt{2(1+\gamma_0)} > 2m$.

Final mass, obtained from inelastic collision of m's, is greater than 2m!

What happens to energy?

initial $\sum \mathcal{E}_0 = mc^2 + \frac{mc^2}{\sqrt{1-v_0^2/c^2}} = mc^2(1+\gamma_0)$

final $\sum \mathcal{E} = \frac{Mc^2}{\sqrt{1-\frac{V^2}{c^2}}}$

$$1-\frac{V^2}{c^2} = 1-\frac{1}{c^2}\left(\frac{v_0\gamma_0}{1+\gamma_0}\right)^2 = 1-\frac{v_0^2}{c^2}\cdot\frac{\gamma_0^2}{(1+\gamma_0)^2} =$$

$$= \left(1-\frac{v_0^2}{c^2} = \frac{1}{\gamma_0^2} \Rightarrow \frac{v_0^2}{c^2} = 1-\frac{1}{\gamma_0^2}\right) = 1-\left(1-\frac{1}{\gamma_0^2}\right)\frac{\gamma_0^2}{(1+\gamma_0)^2} =$$

$$= 1 - \frac{\gamma_0^2 - 1}{(1+\gamma_0)^2} = \frac{1 + 2\gamma_0 + \gamma_0^2 - (\gamma_0^2 - 1)}{(1+\gamma_0)^2} = \frac{2(1+\gamma_0)}{(1+\gamma_0)^2}$$

$$\Rightarrow \sqrt{1 - \frac{v^2}{c^2}} = \sqrt{\frac{2}{1+\gamma_0}}$$

$$\Rightarrow \sum \mathcal{E} = \frac{Mc^2}{\sqrt{\frac{2}{1+\gamma_0}}} = \sqrt{\frac{1+\gamma_0}{2}}\, Mc^2 = \sqrt{\frac{1+\gamma_0}{2}}\, m\cancel{c^2}\sqrt{2(1+\gamma_0)} =$$

$$= mc^2(1+\gamma_0)$$

$\rightarrow$ therefore energy is conserved

$$\sum \mathcal{E}_0 = \sum \mathcal{E}.$$

$\Rightarrow$ The final mass $M > 2m$ because part of kinetic energy of $m$ went into internal (thermal) energy which increased the final total mass.

4-form of Newton's law

4-acceleration $w^\alpha = \frac{du^\alpha}{d\tau} = \frac{d^2x^\alpha}{d\tau^2}$

$$\Rightarrow \frac{dp^\alpha}{d\tau} = m\frac{du^\alpha}{d\tau} = m\frac{d^2x^\alpha}{d\tau^2} = F^\alpha$$

4-force $F^\alpha$ is defined by this equation.

$(\alpha = 1,2,3)$ $\frac{d\vec{p}}{d\tau} = \gamma\frac{d\vec{p}}{dt} = \gamma\vec{F}$ (where $\vec{p} = \frac{m\vec{v}}{\sqrt{1-v^2/c^2}}$)

$\Rightarrow F^{(\alpha=1,2,3)} = \gamma\vec{F}$. $\Rightarrow$ we still have $\boxed{\frac{d\vec{p}}{dt} = \vec{F}}$ for $\alpha=1,2,3$ components.

$(\alpha=0)$ $\frac{dp^0}{d\tau} = \frac{d\mathcal{E}}{c\,d\tau} = \frac{\gamma}{c}\frac{d\mathcal{E}}{dt} = \gamma\frac{\vec{F}\cdot d\vec{r}}{c\,dt} = \gamma\vec{F}\cdot\frac{\vec{v}}{c}$

$$\Rightarrow F^0 = \gamma\vec{F}\cdot\vec{\beta} = \gamma\vec{F}\cdot\frac{\vec{v}}{c}$$

$$\Rightarrow F^\alpha = \gamma\left(\vec{F}\cdot\frac{\vec{v}}{c}, \vec{F}\right).$$

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Massless particles: do not make sense in Newton mechanics.

But do relativistically: $p^2 + m^2c^2 = \frac{\mathcal{E}^2}{c^2} \Rightarrow$

for $m=0$ $p = \frac{\mathcal{E}}{c}$, $\mathcal{E} = cp$.

~~The~~ Photons: $\mathcal{E} = \hbar\omega$, $\vec{p} = \hbar\vec{k}$

$$\Rightarrow p^\alpha = \left(\frac{\hbar\omega}{c}, \hbar\vec{k}\right) = \hbar k^\alpha$$

$$k^\alpha = \left(\frac{\omega}{c}, \vec{k}\right)$$

$$k_\alpha k^\alpha = \frac{\omega^2}{c^2} - k^2 = 0.$$

Note: no rest frame can be associated with massless particle since $ds = 0$.